Event-Driven Multi-Image FPGA Reconfiguration: RTL Architecture and a Preliminary Verification Study

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ABSTRACT

This paper describes the RTL architecture and a preliminary verification study of an autonomous controller that sequences configuration of multiple FPGA images from external serial flash. The platform combines a metadata-driven scheduler, AXI4-Lite control writes, a memory-to-JTAG streaming path, and sticky interrupt handling for target task events. The scheduler advances on configuration and task-completion events, while a runtime limit serves as a safety watchdog. A focused Vivado XSim testbench exercises programmer package definitions, interrupt capture and clearing, and a scheduler AXI-write failure followed by abort handling. The supplied simulation transcript reports 94 package value checks, 12 interrupt scenarios, three scheduler scenarios, and 29 scoreboard checks, completing at 390 ns without a testbench failure. An integrated three-image testbench is also present in the repository and specifies checks for metadata, QSPI transactions, stream beats, JTAG activity, and task-driven phase changes. The available XCRG attempt did not find a functional coverage database; therefore, this study reports directed checks rather than coverage percentages.

Index Terms—FPGA configuration, SystemVerilog, event-driven scheduling, AXI4-Stream, JTAG, RTL verification.

I. INTRODUCTION

FPGAs can be configured through vendor-supported interfaces including JTAG and configuration-memory paths. A multi-image controller must coordinate image metadata, memory reads, streaming transfer, target identification, configuration status, and application-level completion. These responsibilities cross several interfaces and include failure paths that are difficult to assess from an isolated controller test. Vendor documentation defines device configuration behavior, while AMBA specifications define the control and streaming protocols; application-specific sequencing and recovery remain system design concerns [1], [2].

Prior FPGA scheduling research has explored dynamic partial reconfiguration and task preemption, whereas remote JTAG work has studied extending programming paths over optical links [4], [5]. The design studied here takes a different, narrower scope: a full-image, metadata-driven control path that advances between images when the configured workload reports completion. It does not implement partial reconfiguration, task preemption, or a general-purpose operating system scheduler.

This paper makes three bounded contributions. First, it documents an RTL decomposition for autonomous multi-image configuration using a scheduler, control-register hub, memory manager, DMA stream bridge, JTAG controller, and interrupt hub. Second, it describes a self-checking integrated testbench architecture for a three-image QSPI-to-JTAG flow. Third, it reports the results of a focused XSim unit test and distinguishes those checks from unavailable XCRG coverage data. The reported evidence is preliminary: it is simulation-only and does not include FPGA-board measurements, synthesis utilization, or timing closure.

II. PLATFORM ARCHITECTURE

The top-level design connects a time-multiplex scheduler to an AXI4-Lite register bank. Register writes configure the memory manager, DMA controller, and JTAG controller. The selected memory path in the integrated simulation is QSPI; SPI and OSPI controller modules are also present in the RTL. The memory manager returns metadata and image bytes over a stream interface. During metadata acquisition, the scheduler consumes the stream directly; image data is routed through the DMA bridge to the JTAG controller.

Fig. 1. Control and configuration data paths in the RTL platform.

CONTROL	Scheduler → AXI4-Lite register hub → memory / DMA / JTAG
DATA	QSPI NOR → memory manager → DMA stream bridge → JTAG → target TAP
FEEDBACK	Target configuration status and task events → synchronizers / IRQ hub → scheduler
PACKAGE	programmer_pkg supplies shared parameters, encodings, packed types, and CRC helper

The scheduler reads a metadata table containing one four-word record per image: flash offset, bitstream length in bits, expected IDCODE, and maximum task duration. It writes configuration values through a single-outstanding AXI4-Lite write master, then waits for the memory, DMA, and JTAG engines to complete. It requests target configuration status before enabling the fabric. After release, a sticky target-task-done interrupt advances the image phase; a task-error interrupt or watchdog expiry leads to the abort sequence. Durations are runtime limits, not fixed phase timers.

The DMA controller is a single-entry AXI4-Stream bridge. It checks nonzero byte-aligned transfer lengths, contiguous byte enables, expected final-beat framing, and the declared bit count. A registered output beat provides one-entry backpressure decoupling. The JTAG controller drives the target TAP and supports IDCODE capture, instruction scans, data shifts, and abort recovery. These interfaces follow the protocol roles of AXI4-Lite, AXI4-Stream, and IEEE 1149.1 JTAG [2], [3].

III. VERIFICATION METHOD

A. Integrated testbench structure

The repository contains a procedural SystemVerilog top-level testbench, a behavioral multi-image QSPI NOR model, and a behavioral JTAG target model. The testbench initializes three distinct nine-byte images and twelve metadata words, starts the scheduler, and models target configuration status and task-completion events. It checks metadata transfer counts, QSPI opcodes and addresses, memory-to-DMA and DMA-to-JTAG beat contents, JTAG TDI bits, IDCODE and instruction selection, status handshakes, task timing, interrupt acknowledgement, and final scheduler status. The design uses a 10 MHz simulation clock and requests three images with distinct task runtimes of 45, 120, and 15 microseconds.

The integrated testbench is a directed nominal-flow test, not a UVM environment. Its monitors and scoreboard use procedural checks that call $fatal on mismatches. The repository snapshot does not contain the UVM driver, sequence, scoreboard, or four-test regression suite described in the supplied project report. Accordingly, UVM and those error-injection results are not treated as executed evidence here.

B. Focused module test

A separate testbench instantiates programmer_pkg-dependent checks, the interrupt controller, and the scheduler with an AXI-Lite response model. Package checks cover the declared memory, trigger, command, response, error, interrupt, DMA, controller-state, and TAP-state encodings, as well as the CRC update function. Interrupt scenarios exercise reset, registered enable configuration, rising-edge capture, sticky status, masking, clearing, held-high sources, simultaneous sources, and clear priority. The scheduler test injects an AXI write error, services the subsequent abort writes, and checks that the controller reaches failure with the target fabric disabled.

This focused test is also directed. Its totals are scoreboard counters and scenario checks; they are not covergroup hit counts. Although

SystemVerilog covergroups were added to the testbench after the supplied passing transcript, the transcript used for this paper predates that change. The subsequent XCRG invocation reported that no functional coverage database was present. Therefore, no XCRG percentage or bin-closure result is reported.

IV. PRELIMINARY RESULTS

The supplied Vivado XSim transcript for the focused testbench shows successful compilation and elaboration, followed by completion at 390 ns without a testbench failure. The testbench printed the following directed-check totals. These counts describe its checks and do not estimate the fraction of RTL behavior exercised.

TABLE I. FOCUSED XSIM TESTBENCH COUNTS REPORTED BY THE SUPPLIED RUN

Target	Reported checks	Interpretation
programmer_pkg	94	Package value/function checks
interrupt_controller	12	Directed scenarios
time_multiplex_scheduler	3	Directed scenarios
Combined scoreboard	29	Individual pass/fail checks
Simulation	390 ns	Reached $finish; no TB failure

The integrated testbench source defines stronger end-to-end nominal checks: twelve metadata handshakes, four QSPI reads (metadata plus three images), three configuration-status handshakes, three task-done events, twelve JTAG completion events, sixty-four AXI-Lite writes, and seventy-two shifted image bits per phase. These values are testbench expectations encoded as pass/fail checks. The repository snapshot includes the testbench but not a corresponding successful XSim transcript, so this paper does not present them as measured run results. A project report separately records a 655.75 microsecond nominal completion, but the underlying run log and UVM sources are not present in the checked-in snapshot; that timing is omitted from the measured results.

V. DISCUSSION AND LIMITATIONS

The focused run demonstrates that the package checks, interrupt edge and sticky-status scenarios, and one scheduler write-error recovery path complete under the supplied XSim setup. It is useful as a fast unit-level regression, while the integrated testbench provides a concrete structure for checking interactions across flash, stream, JTAG, and target-event boundaries. The two levels should remain distinct: the focused run does not establish end-to-end operation, and the integrated testbench source alone does not establish a passing simulation.

Several evidence gaps remain before this work is ready for a full research submission. The XCRG report could not be generated because the supplied run produced no functional-coverage database. The repository contains no UVM sources or automated four-test regression script, and the DMA RTL still uses $countones() rather than the iterative keep_to_bits() fix described in the project report. No synthesis, resource-utilization, maximum-frequency, physical-board, or fault-injection measurements are available in this snapshot. These items should be reconciled and rerun before making coverage-closure, portability, or hardware-readiness claims.

A stronger evaluation should archive exact source revisions and simulator logs; rerun the integrated nominal test and each negative test; generate an XCRG database from the final source; report the covergroup model and bin definitions; and add synthesis and timing data for a named FPGA part. If the intended contribution is the documented simulator-portability fix or UVM-based fault-recovery methodology, those source changes and regression artifacts should be checked into the same revision used for submission.

VI. CONCLUSION

This paper documents an RTL platform for event-driven sequencing of multiple full FPGA images and a preliminary layered verification approach. The architecture combines metadata parsing, AXI-Lite configuration, QSPI-backed streaming, DMA flow control, JTAG operations, and sticky target-event interrupts. The available XSim evidence is a focused directed test that reports 94 package checks, 12 interrupt scenarios, three scheduler scenarios, and 29 scoreboard checks, completing at 390 ns. Because the XCRG database and full-regression artifacts are unavailable, the results are presented as preliminary checks rather than coverage closure. The next step is to make the source, regression, and coverage database mutually reproducible before evaluating this design as a conference-ready verification methodology.

REFERENCES

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